KiCad 101 to Advanced AI Co-Pilot

EP.2 : PCB Design



Uncle Engineer

ลุงวิศวกร สอนคำนวณ

PCB Design Overview

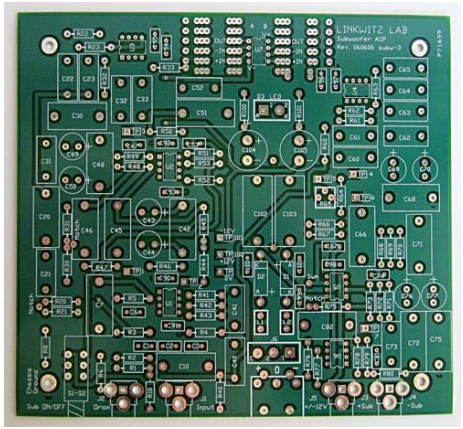
อยากทำ PCB อย่างนั้นเหรอ???????????! ! !

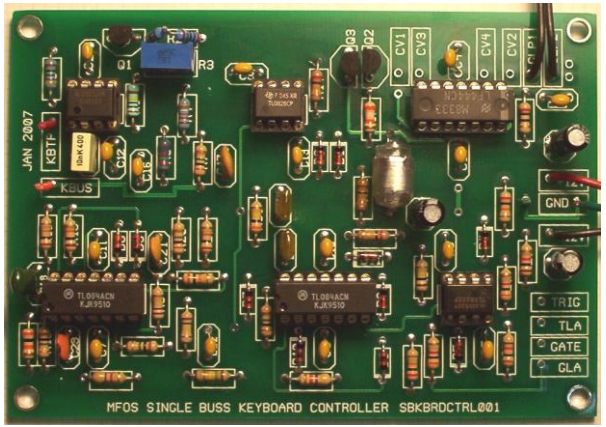
PCB เป็นเพียงชุดสายไฟที่เชื่อมต่อส่วนประกอบต่างๆ แต่ทำไมเราต้องใส่ใจในการออกแบบ

- The wires have restricted dimensionality
- The wires are very thin

  So high resistance (as conductors go)

- The board needs to include holes (or pads) for the devices.
- You can’t easily change things once you build it.



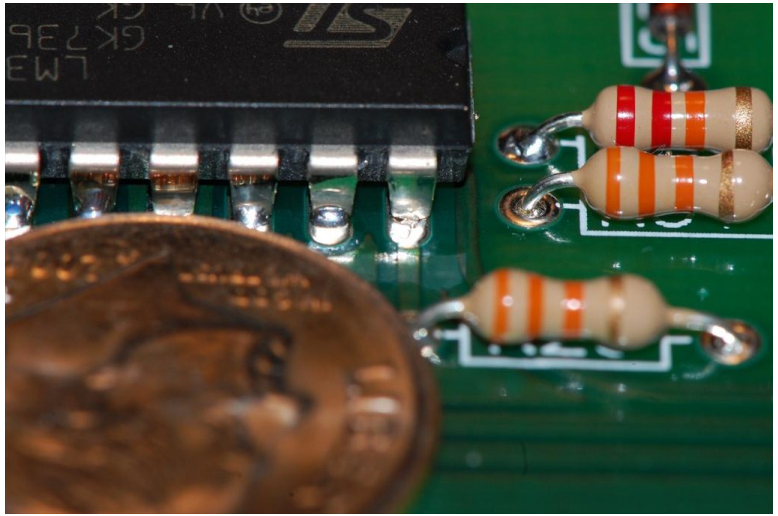


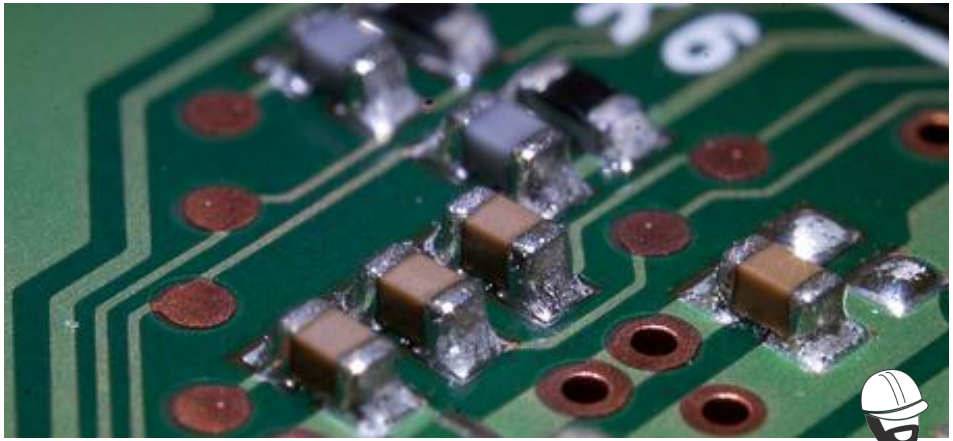
คำศัพท์พื้นฐาน

- The wires you are laying out are called “traces” or “tracks”
- Inside of a given “layer” tracks which cross are electrically connected.

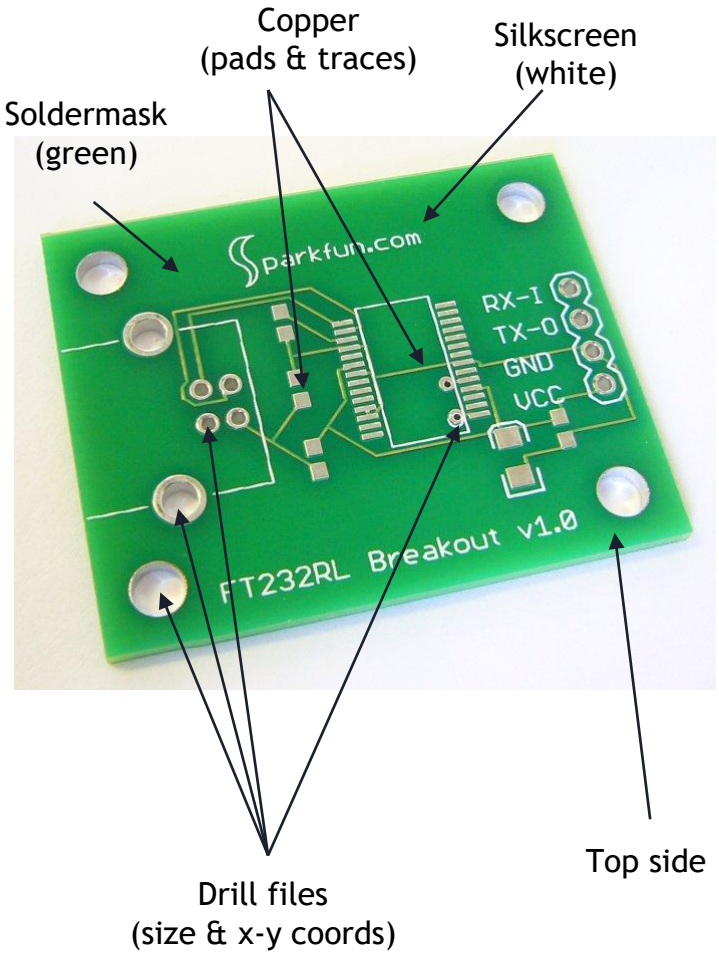
  If you have traces on both sides of the board, you are said to have to layers.
- Through-hole: Having pins designed to be inserted into holes and soldered to pads on a printed board.

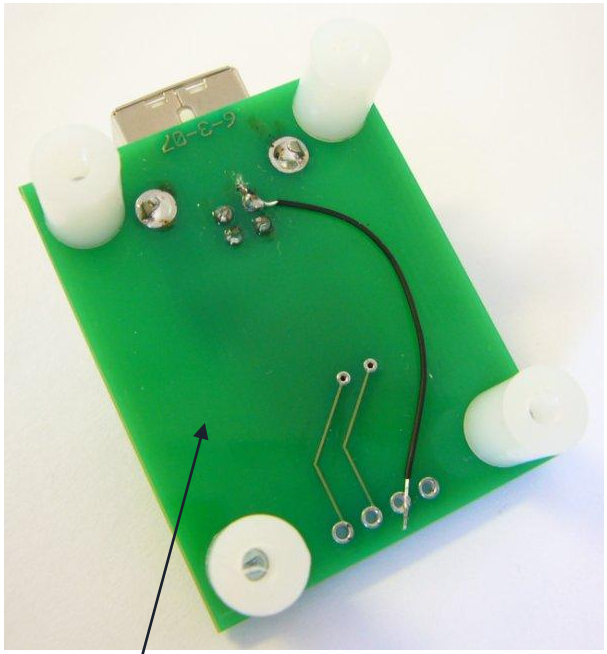
  Contrast with surface mount where device goes on top.



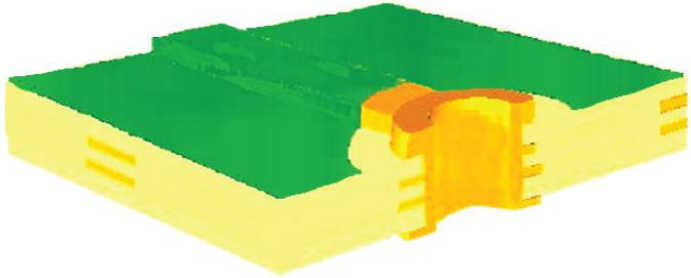


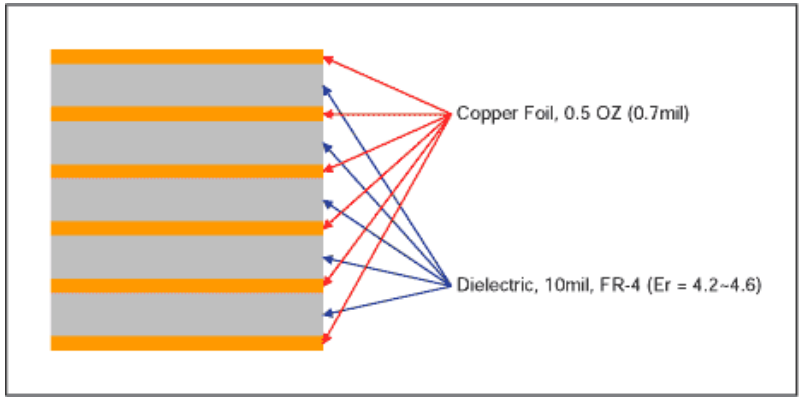
คำศัพท์พื้นฐาน


Copper
(pads & traces)
Silkscreen
(white)
Soldermask
(green)
Sparkfun.com
RX-I
TX-O
GND
VCC
FT232RL Breakout v1.0
Drill files
(size & x-y coords)
Top side

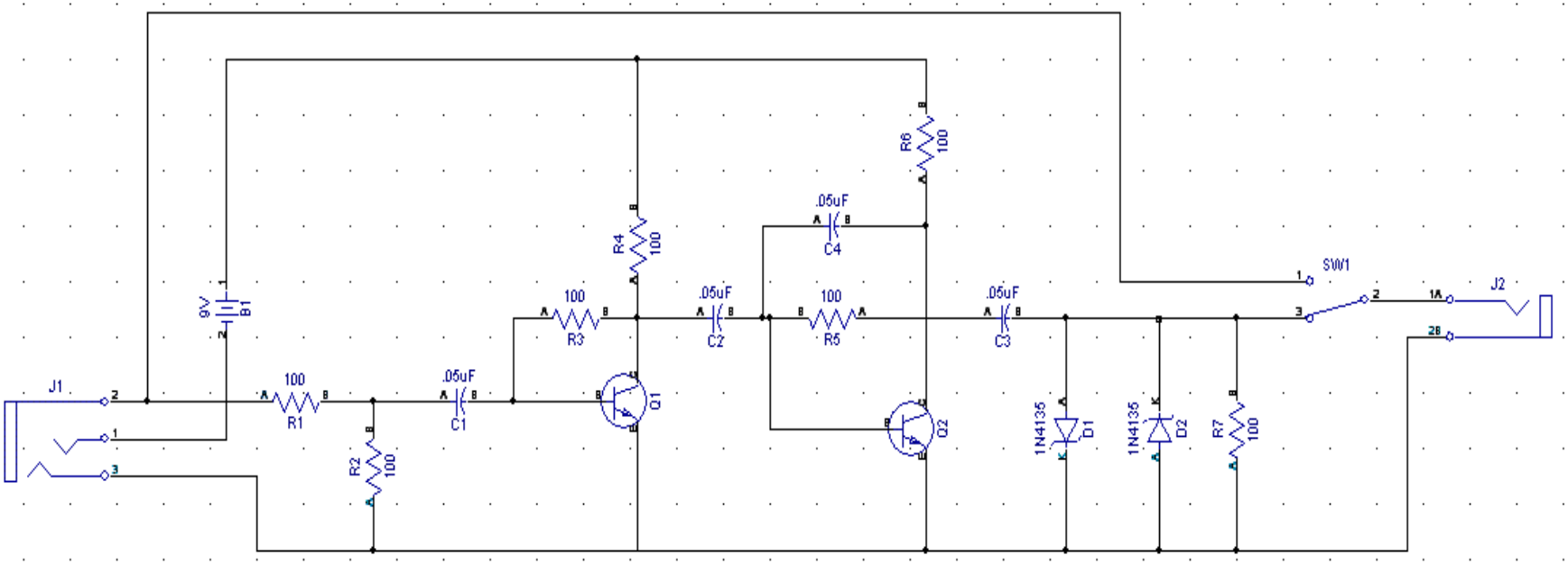

Bottom side

The cross-section of a PCB shows its layers





Step 1: Schematic

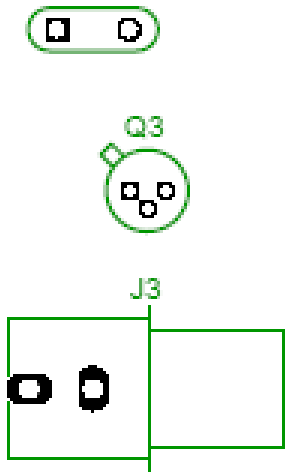


ทำไมต้องใช้ schematic

โดยทั่วไปจะวาดให้อ่านง่ายเป็นลักษณะภาพร่างของบนกระดาษ
สามารถค้นหาและแก้ไขจุดบกพร่องได้ง่ายกว่าเค้าโครงแบบ PCB

Step 2: Patterns

- Once you know what it is you want to build, you need to figure out how to lay it out on the board.
    - You need to know how big each piece is, and where the holes need to be placed.
- Each device has a pattern which shows exactly that.
    - You will occasionally need to create a pattern.



Step 3: Placement

- You need to place the patterns on the board.
  - You need to not overlap them so that the components can actually fit on the board.
  - You want to leave room for the traces to connect everything.
- This is *very* much an art form.
  - In fact you will find people who rant about “unprofessional” placements.
- Some tools will do this for you.



Step 4: Routing

- A route (or net) is a connection between devices.
  - It may consist of multiple traces

- There are design rules which include:
  - Minimum trace width
  - Minimum spacing between traces and holes
  - Minimum spacing between holes and holes.

- These rules will vary by manufacturer.
  - Even better, ***units*** will vary by manufacturer!



Issues of measure

PCB land uses some interesting terminology.

A “thou” is a thousandth of an inch.

A “mm” is a millimeter

A “mil” is a thousandth of an inch.

Thou is generally preferred over mill to avoid confusion, but most tools/vendors use mil.



Trace width

- In general most PCB manufacturers seem to have trace-width minimums of 6-10 thous.
  - Most are willing to go smaller for a price.
- A rule of thumb is to use a 50 thou minimum for power/ground and 15-25 for everything else.
  - This is to drop the resistance of the traces.
  - In general you are worried about heat dissipation
- There are lots of guidelines for width/power but in general you are looking at:
  - A 10cm trace needs to be 10 thou wide if it will carry 1 amp.
  - 5 amps at 10cm would require 110 thou.



Trace width continued

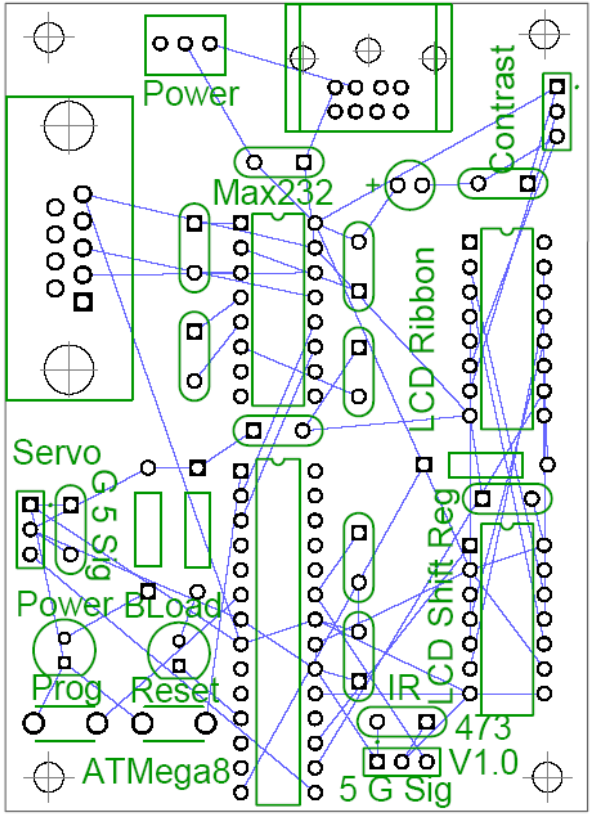
- The *problem* with wide traces is that they are hard to route.
  - In particular you might wish to go between pins of a device.
- One solution is to be wide normally and "neck down" when you have to.
  - This is more reasonable than you think.
    - Think resistors in series.





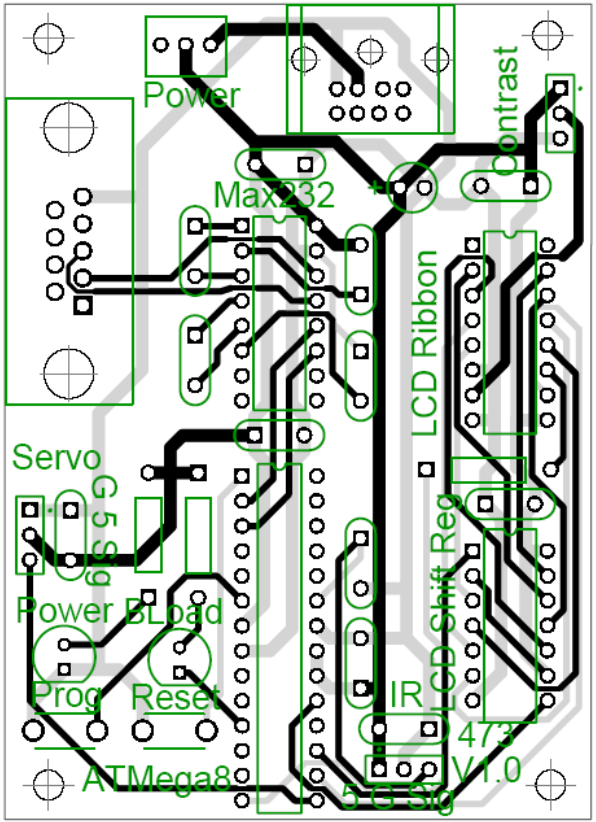
Rat’s nest.

A rat’s nest shows the placement of the devices **and** the connections but not the routing



Routing for real

- You can use an auto-router to route your traces
  - Some people hate these as the design will be “ugly”
  - Saves a lot of time.
  - Oddly, not as good as a person can do.
    - But much faster.



Step 5: Via

Sometimes you need to connect two traces on two different layers.

To do this we use a via.

It is just a plated through hole

Generally smaller than a through hole for a part.



Step 6: Clearances

Again there will be space between the traces, plated holes and each other.

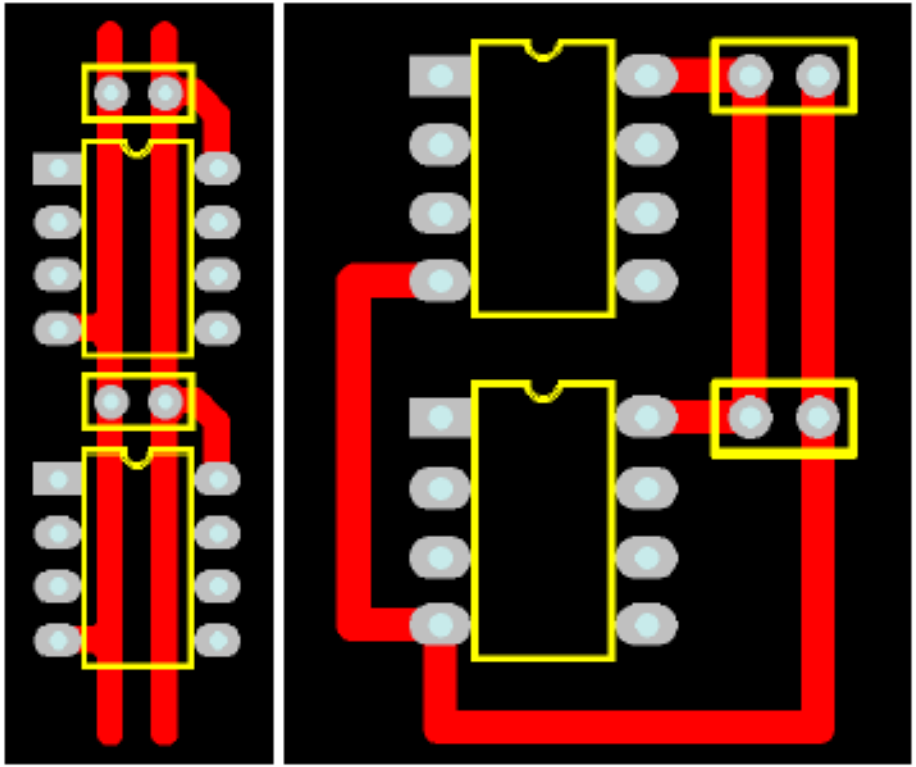
You need to meet the requirement of the manufacturer.

15 thou is a good idea

Often you can drop to 6 or 10.

For high-power systems there are rules about this stuff.





An example of GOOD power routing (Left) and BAD power routing (Right)

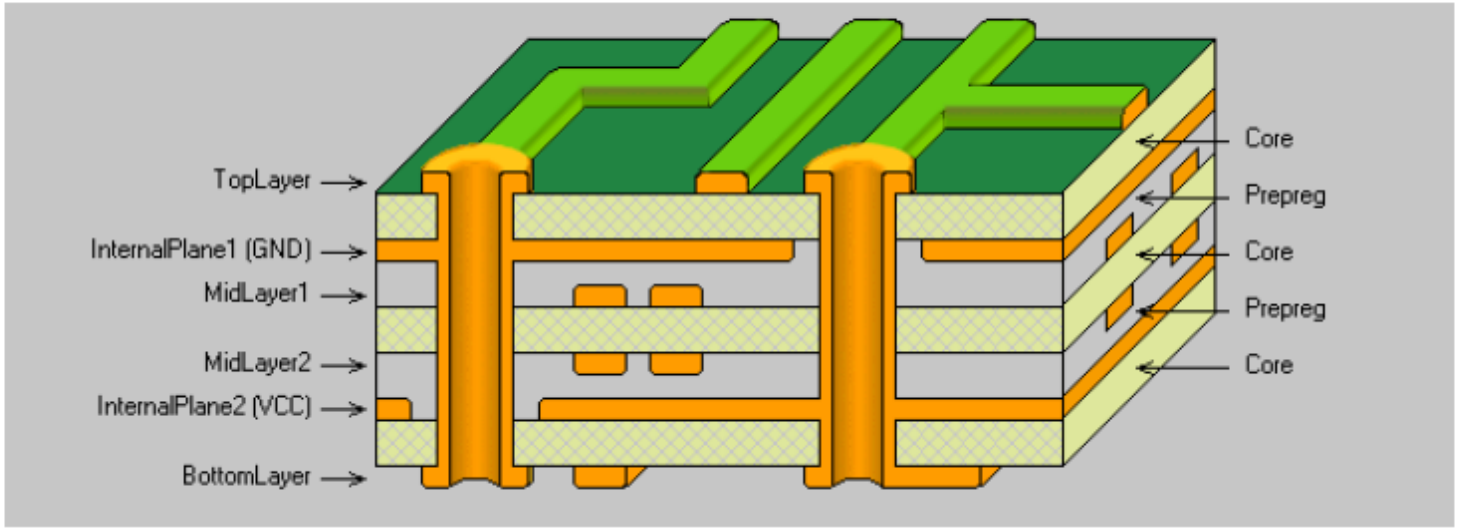
Step 7 : อื่น ๆ

- **Silkscreen**
  - Use this to label parts to insert, directions parts go, names user might want (on/off for example)
- **Solder mask** or **solder resist**
  - a lacquer-like layer of polymer that provides a permanent protective coating for the copper traces of a (PCB) and prevents solder from bridging between conductors, thereby preventing short circuits.
  - Solder mask is traditionally green but is now available in many colors.



Step 8: Putting it all together

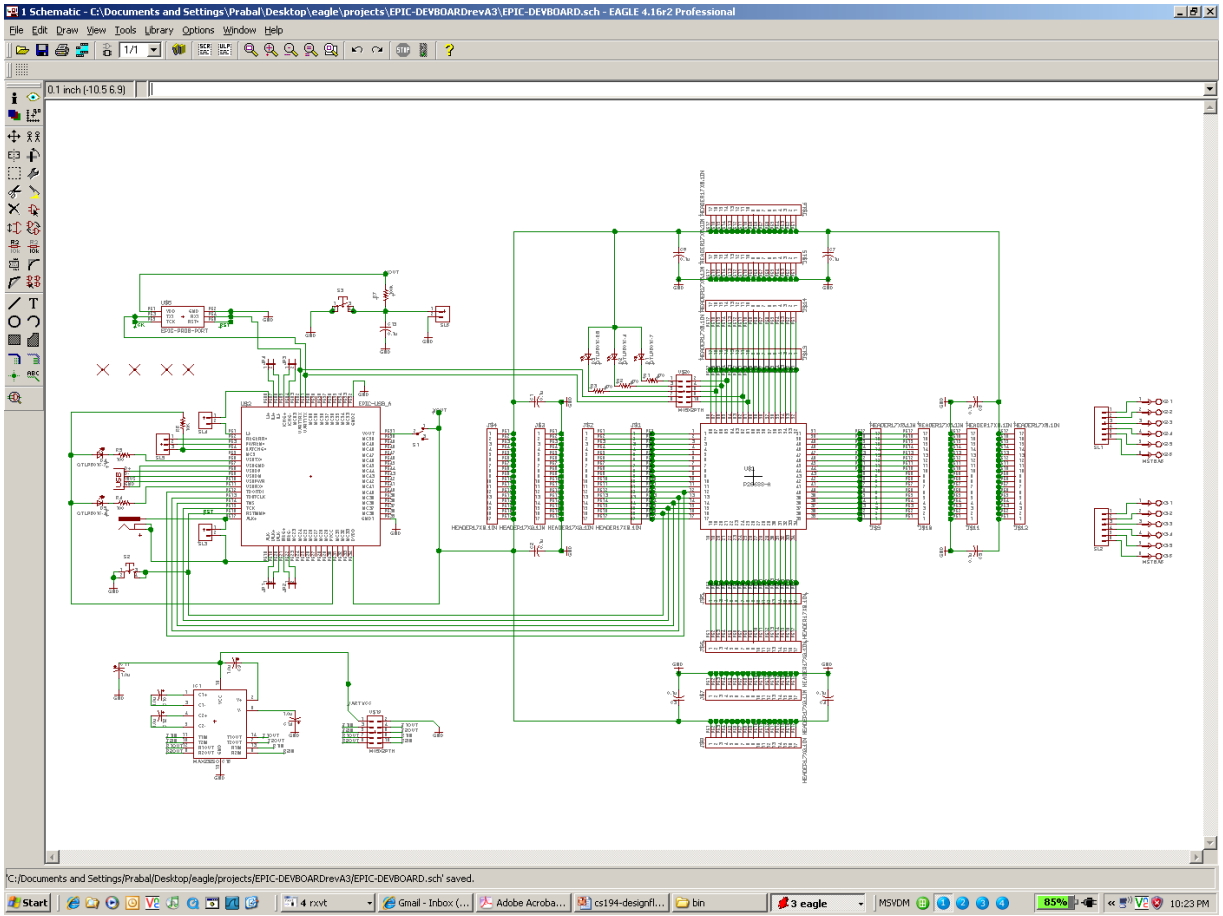
Multi layer Design



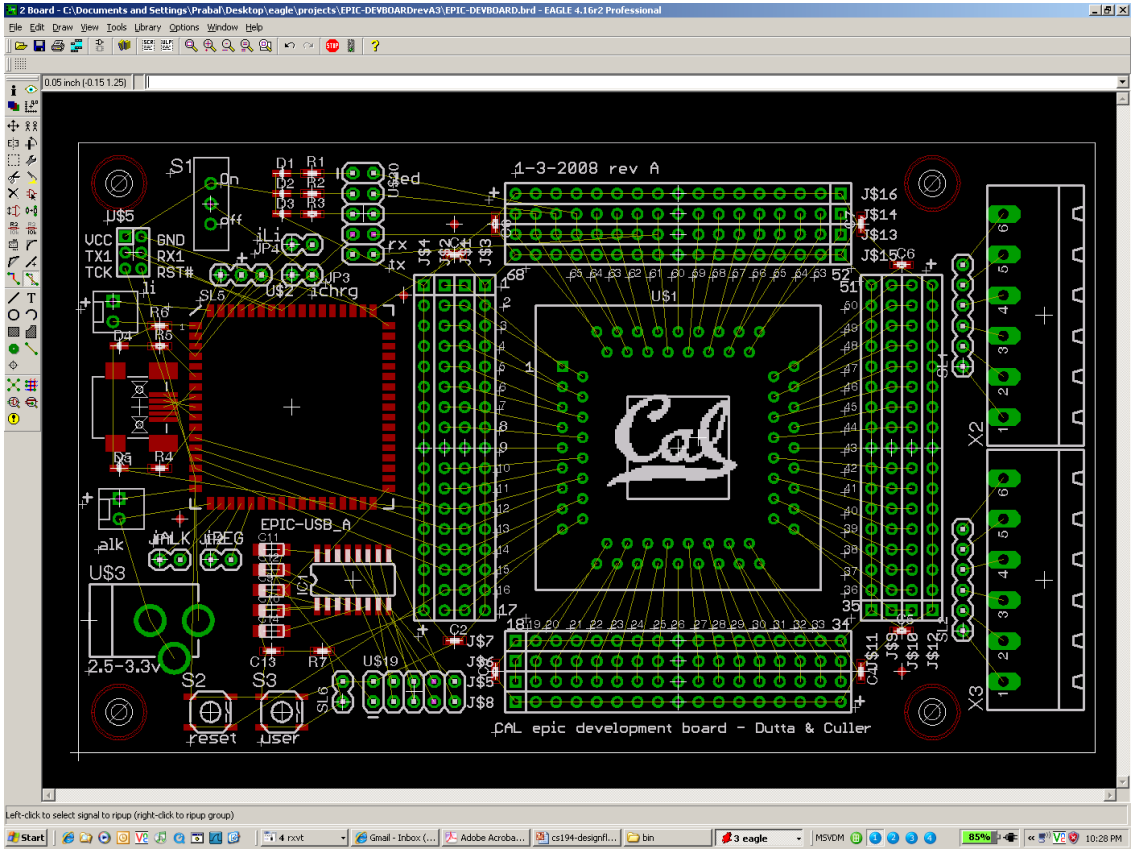
Typical 6 Layer PCB Construction

Electronic Design Automation
&
Computer Aid Design

The schematic captures the logical circuit design



Floor planning captures the desired part locations



The auto-router places tracks on the board, saving time

